Lab 08 — Screen Customers with Lambda, Map and Filter

Python Programming for Finance · TGS-2026065049 · LO4

Scenario

A retail bank wants to transform customer income into affordability ratios and filter applications for manual review.

Goal

A readable transformation pipeline plus an equivalent named-function implementation.

Tools

Python, csv, unittest

Step-by-step

1. Load applicants.csv and count the input records.

2. Use map with a lambda to add monthly_income.

3. Use filter to select high debt-to-income records.

```
python starter.py
```

4. Rewrite the risk rule as a named function.

5. Confirm both versions return identical IDs.

```
python -m unittest test_solution.py
```

6. Explain which implementation is easier to audit.

Acceptance criteria

Lambda and named-function pipelines select the same applicant IDs, and counts reconcile to the input.

Evidence

Capture the executed output, validation result, generated file or chart, and a one-sentence finance interpretation.